

Component Modeling & Validation

Closed-form latency equations from real-silicon Metis measurements — calibrated, gated, and honestly bounded.

2026-06-05 decode-calibrated software-only · Mac branch phase-1

Calibration-scope declaration. Phase 1 is a *decode-calibrated* model. The whole prefill path is analytic and **unvalidated**; the only hard gate is the 8B decode hold-out ($\leq 25\%$). Per-op equation gates follow ADR-0006 (median $\leq 10\%$, p95 $\leq 20\%$). Every number here regenerates from committed JSON.

GATE SUMMARY

MODULE	COMPONENT	EQUATION	RESULT	GATE
M1	CIM tile	$N < 2048: 2\text{MKN}/G_{\text{eff}}(N) \cdot \text{else}$ $M \cdot n_{\text{tiles}} \cdot T_{\text{tile}}$	median 3.4%, p95 8.9%	PASS
M2	PCIe / DMA	floor + bytes/BW (911 μ s, 3.9 GB/s)	positive, monotonic; boundary recorded	PASS
M2	LPDDR5	analytic eff-BW (Ramulator2 $\rightarrow$ Phase 2)	24.2 / 51.2 GB/s (47%)	PASS
M4	GPU (Mali) attn offload	$\text{attn} = 27.742 + 0.44208 \cdot \text{kv } \mu\text{s}$	median 0.6%, p95 1.1%	PASS
M4	CPU (A76)	softmax $a+b \cdot \text{kv}$; others constants	median 0.3%, p95 1.8%	PASS
M4	NPU	—	blocked on issue #13	PENDING
M5	trace	deterministic (Phase 0.1 oracle)	4 models, 0 orphans	PASS
M7	energy	spec (15 TOPS/W, pJ/bit, A76 W)	memory-dominated, $\pm 20\%$ robust (0 flips)	PASS
M3/M6	engine / scheduler	—	contract only (Phase 2); conversion-op + knee tunables	CONTRACT

MODULE	COMPONENT	EQUATION	RESULT	GATE
E2E	recompose	$\text{tok_s} = \text{BW_eff} / \text{weight_bytes}$ (1b/3b→8b)	8B 9.5% (2.44 vs 2.7)	PASS

HEADLINE RESULTS

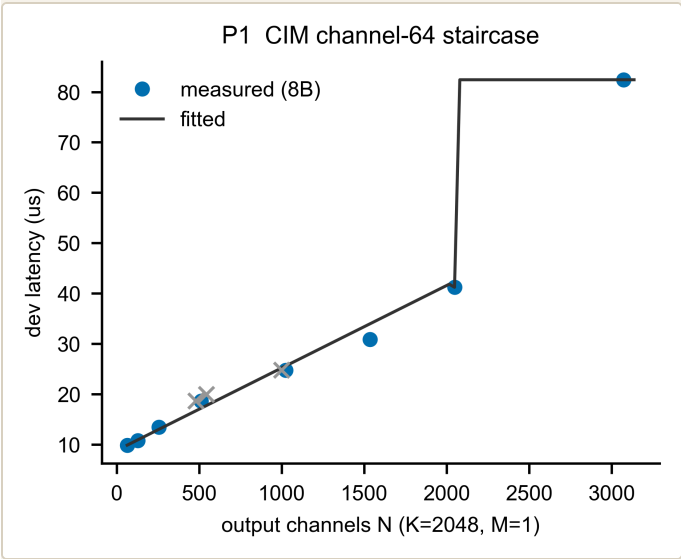
CIM excels at weight-stationary matmul. The decode-GEMV throughput curve $G_{\text{eff}}(N)$ fits the channel-64 staircase to **3.4% median** (8.9% p95). Full/multi-tile projections follow $n_{\text{tiles}} \cdot T_{\text{tile}}$ ($T_{\text{tile}} = 41.21 \mu\text{s}$), reproducing the padded-tile measurements (incl. Qwen, no restore).

Attention must offload. Composed CIM attention is **31–46 ms/token** (C4) vs Mali GPU-native tens-to-hundreds of μs — ≈ 2 orders (96–370 $\times$), the empirical basis for "design around CIM, offload attention".

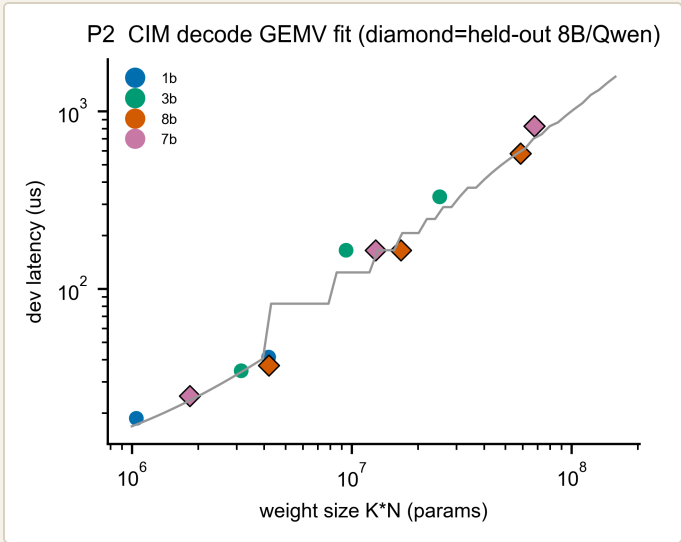
GQA narrows underfill the crossbar. Wide-K narrow-N kv-projections are over-predicted by $G_{\text{eff}} - 8\text{B kv} + 39\%$ — a CIM-centric finding reported separately, not buried in the average.

Decode is the memory wall. The recompose backbone $\text{tok_s} = \text{BW_eff}/\text{weight_bytes}$ (fit 1B+3B) predicts held-out 8B to **9.5%**. Spec energy: 8B decode is memory-dominated (CIM 1.0 mJ vs DRAM 240 mJ/token), robust to $\pm 20\%$.

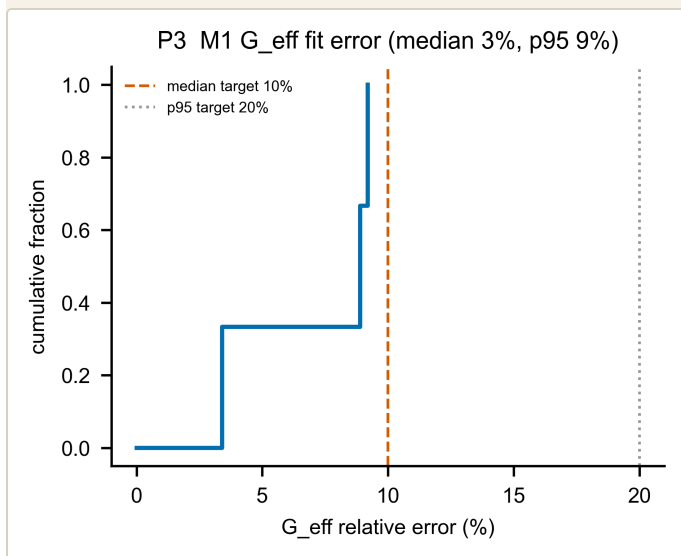
FIGURES



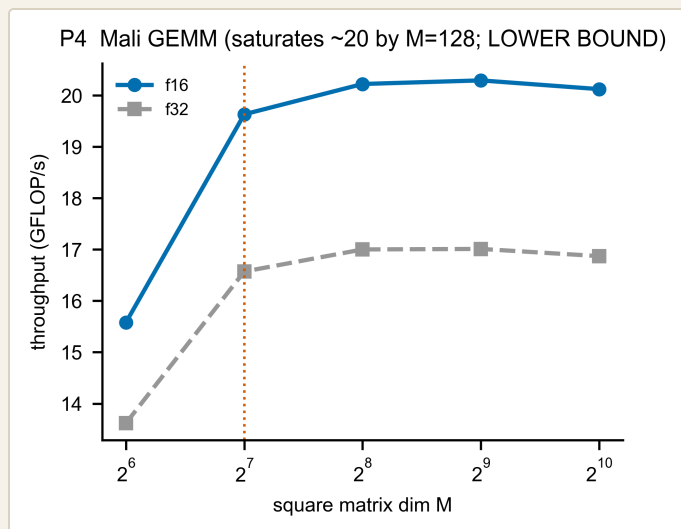
P1 – CIM channel-64 staircase: measured (8B) vs fitted G_{eff} ; off-64 probes (x). Tile jump at $N > 2048$.



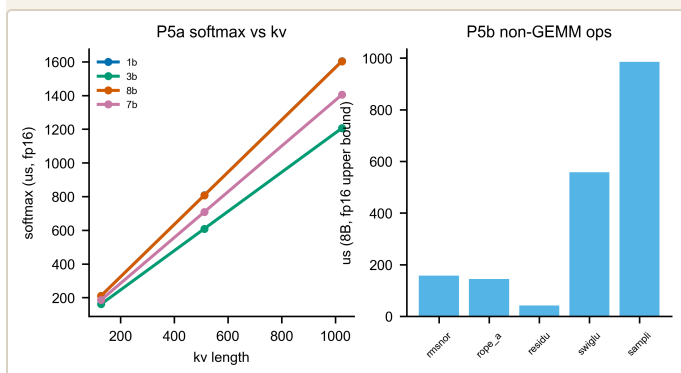
P2 – CIM decode-GEMV fit across all model shapes; diamonds = held-out 8B/Qwen projections.



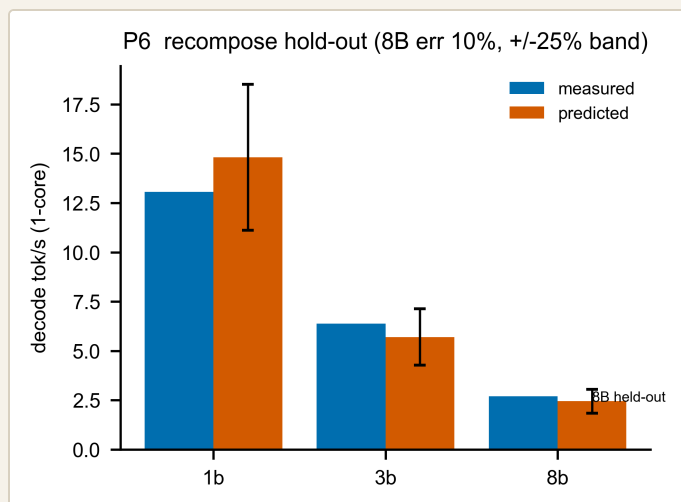
P3 – M1 G_eff fit error vs the 10% median / 20% p95 targets.



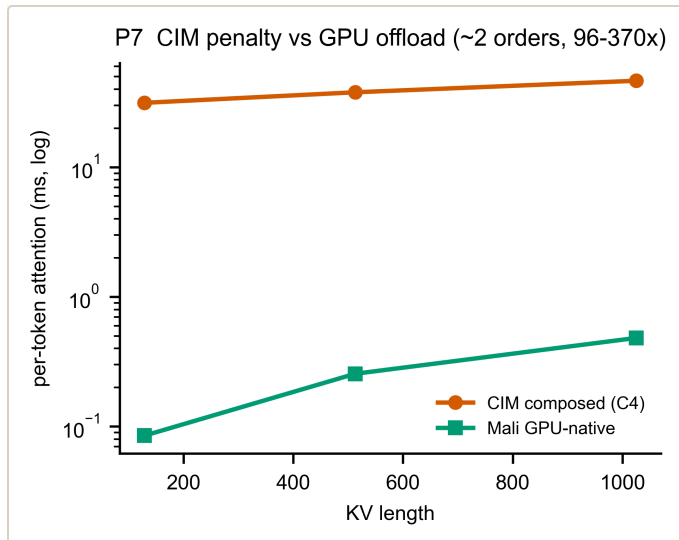
P4 – Mali GEMM throughput vs size (f16/f32); saturates ~20 GFLOP/s by M=128. Absolute = lower bound.



P5 – CPU support ops: softmax linear-in-kv (a) and non-GEMM constants (b), fp16 upper bound.



P6 – End-to-end recompose: measured vs predicted decode tok/s; 8B held-out within $\pm 25\%$.



P7 – CIM attention penalty (C4) vs Mali GPU-native:
~2 orders (96–370×) → offload attention.

GAPS & DEFERRALS

NPU (M4) — issue #13, not collected. **Conversion-op cost (M6)** — ADR-0004 Phase-0.2 calibration never done; the headline mixed-precision contribution has no cost basis yet (tracked tunable + measurement gap). **Prefill** — decode-calibrated only; vendor TTFT implies ~4.1 TOPS prefill GEMM throughput, unmeasured (proj M≥512 device-fail). **kv_cache / embedding** — analytic, unvalidated. **Ramulator2** — deferred to Phase 2 (analytic LPDDR5 ships now). **Phase-2 watch-items** — kv_append vs BW double-count; attention heads×layers rollup.

Build artifact · regenerate: `./.venv/bin/python tools/analysis/fit_m1_cim.py · fit_m2.py · fit_m4_gpu.py · fit_m4_cpu.py · recompose_e2e.py · tools/plotting/phase1_figs.py · tools/report/build_phase1_report.py`